



SEGMENTATION AND PAGING IN OPERATING SYSTEMS

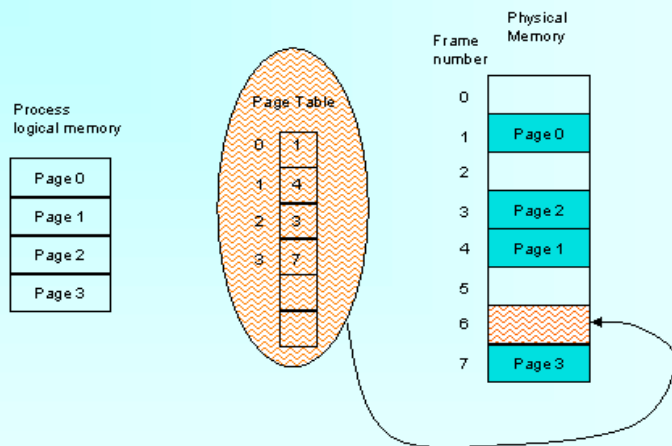
Exploring the fundamental concepts of segmentation and paging in memory management.

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What is Paging?

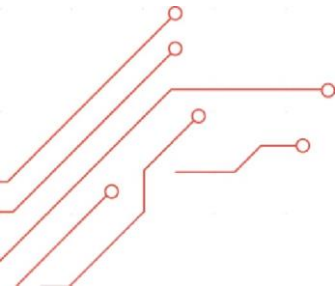
- Paging is a memory management technique in which memory is divided into fixed size pages.
- Paging is used for faster access to data.
- It is a logical concept.

Example of Paging

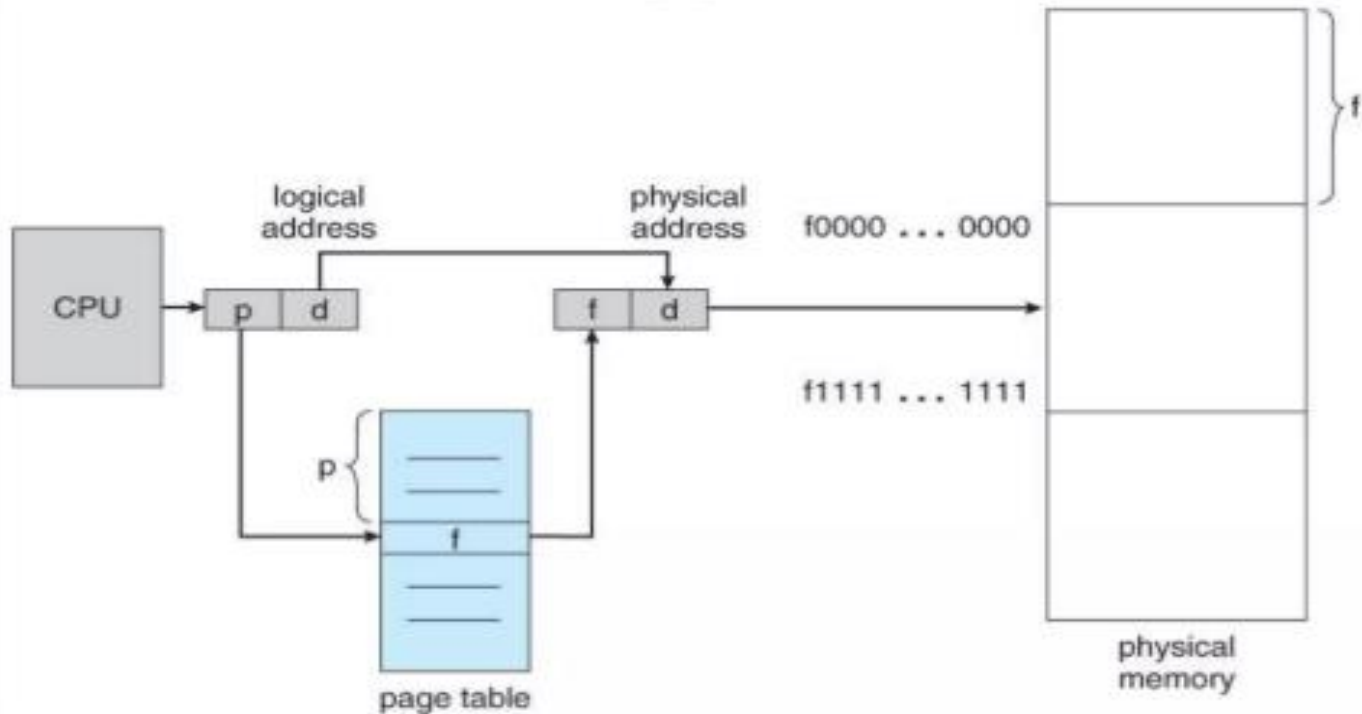


Why we use paging ?

- Paging is a memory management technique.
- Paging allows the physical address space of the process to be non contiguous.
- Paging is use for faster access to data.
- It is a logical concept.



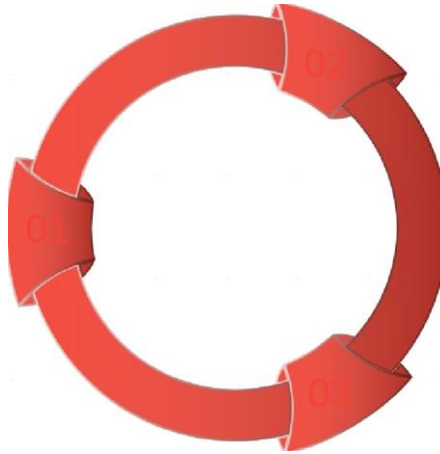
Paging Hardware



UNDERSTANDING PAGING IN MEMORY

PAGE SIZE DEFINITION

The page size is set at 4KB, which is a standard size used in many systems for memory management. This size determines how data is divided into pages for efficient memory use.



MAPPING PAGES TO FRAMES

In this example, Page 2 is mapped to Frame 5. This mapping is crucial for the operating system to keep track of where data is stored in physical memory.

ADDRESS CALCULATION

The formula for calculating the physical address involves multiplying the frame number by the page size and adding the page offset. The address for Page 2 in Frame 5 is calculated as $(5 \cdot 4096) + \text{Offset}$.

Advantages and disadvantages of paging

- **Advantages :**

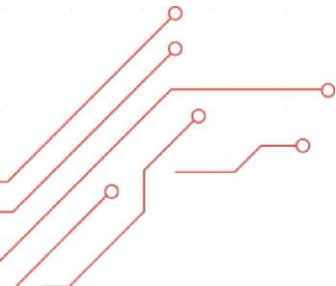
- no external fragmentation.
- Simply memory management algorithm.
- Swapping is easy(equal sized pages and page frame).

- **Disadvantages:**

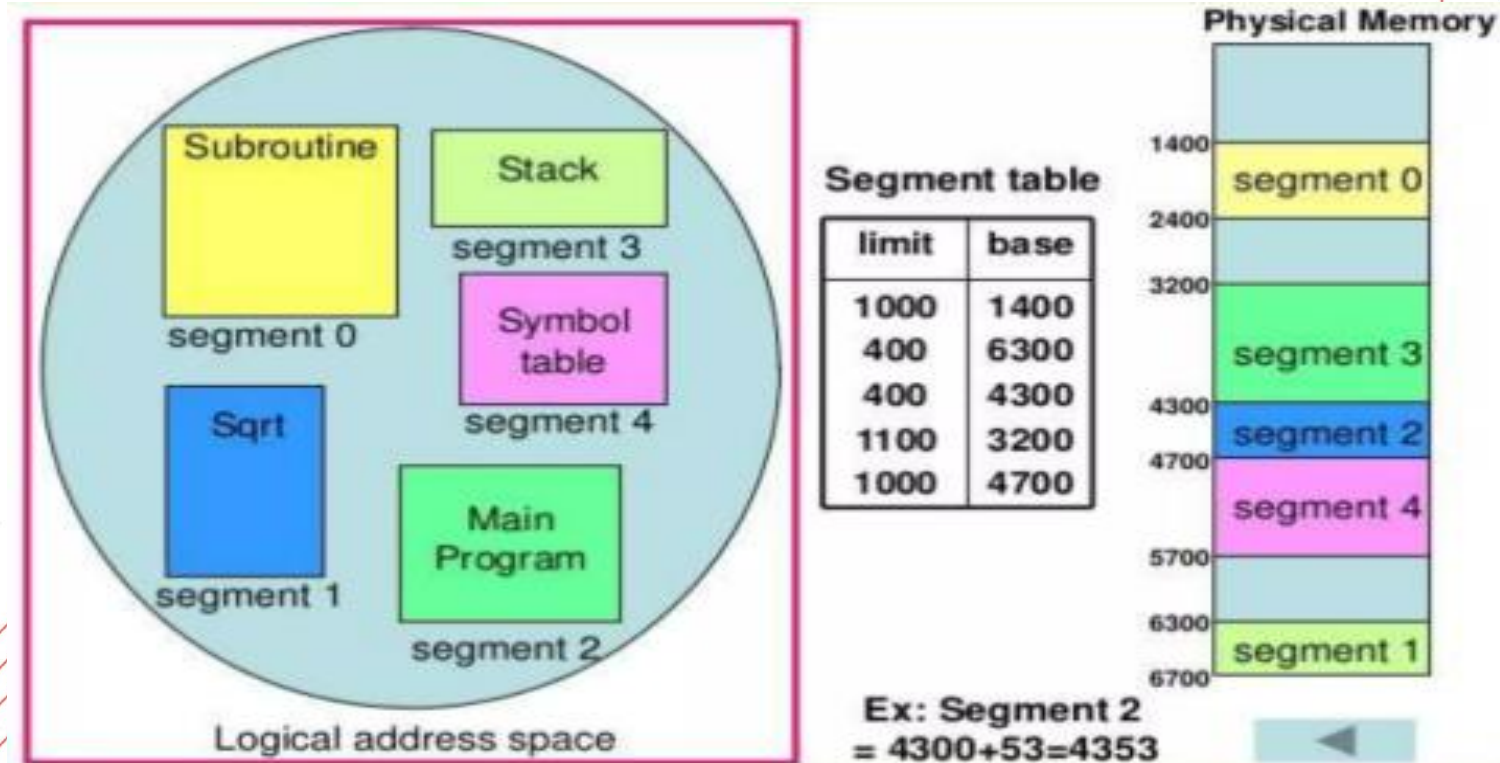
- Internal fragmentation.
- Page tables may consume more memory.

What is Segmentation ?

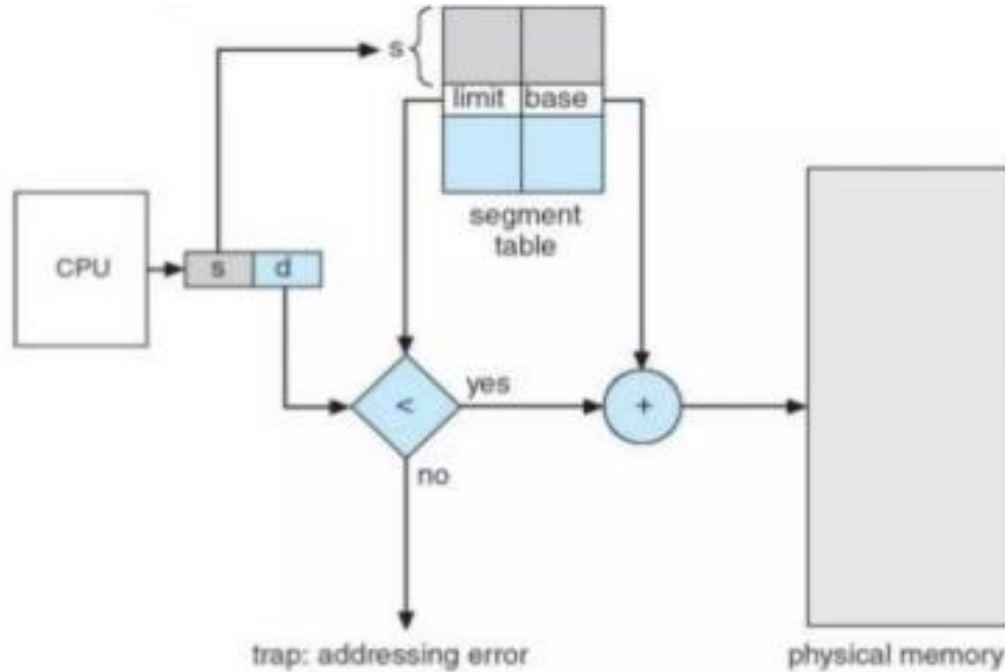
- Segmentation is one of the most common ways to achieve memory protection.
- The user will view as a combination of segments.
- Each memory segment is associated with a specific
- length and set of permissions.



Example of Segmentation



Segmentation Hardware



Advantages and disadvantages of segmentation

- **Advantages :**
 - no internal fragmentation.
 - Segment tables consume less memory than page.
 - Lends itself to sharing data among processes.
 - Lends itself to protection.
- **Disadvantages:**
 - Costly memory management algorithm.

SEGMENTATION VS PAGING OVERVIEW

SEGMENTATION IS A LOGICAL DIVISION

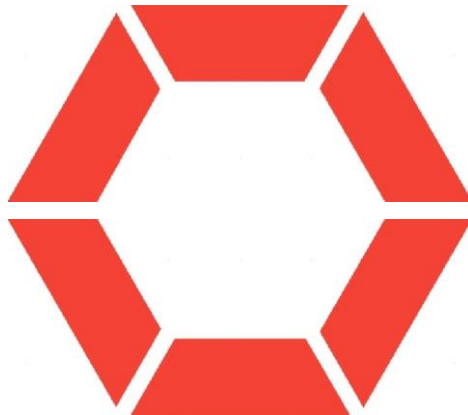
In segmentation, memory is divided into variable-sized segments based on the logical structure of the program, such as functions or data types, enhancing modularity.

PAGING SIMPLIFIES MEMORY MANAGEMENT

Paging simplifies memory management by allowing the operating system to manage memory in fixed-size units, reducing complexity in allocation and deallocation.

SEGMENTATION ENHANCES MODULARITY

Due to its logical structure, segmentation allows programs to be more modular and facilitates easier sharing and protection of segments between processes.



PAGING USES FIXED-SIZE BLOCKS

Paging divides memory into equal-sized blocks called pages. This technique allows for efficient memory allocation and eliminates the need for contiguous memory allocation.

EXTERNAL FRAGMENTATION IN SEGMENTATION

Segmentation can lead to external **fragmentation**, where free memory is split into small non-contiguous blocks, making it difficult to allocate larger segments.

INTERNAL FRAGMENTATION IN PAGING

Paging may cause internal fragmentation, where allocated pages may have unused space if the data does not completely fill the page, leading to wasted memory.

OVERVIEW OF MEMORY MANAGEMENT



SEGMENTATION ENHANCES PROGRAM STRUCTURE

Segmentation divides a program into different segments based on logical divisions like functions or data, improving organization and management.



PAGING OPTIMIZES MEMORY USAGE

Paging allows the OS to use physical memory more efficiently by dividing it into fixed-size pages, reducing fragmentation and improving allocation.



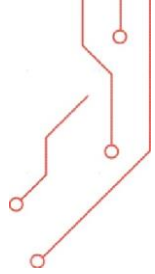
MEMORY EFFICIENCY IS CRUCIAL

Both segmentation and paging are vital techniques for efficient memory management in modern operating systems, ensuring optimal resource utilization.



MODERN OS RELIES ON BOTH

The combination of segmentation and paging techniques is fundamental in modern operating systems for effective memory handling and performance.



Thank You